

Challenge (Habanero)

- Q1.** Simplify $2^3 \cdot 2^4$
(A) 2^7
(B) 2^6
(C) 2^5
(D) 2^3
(E) 2^2
- Q2.** A recipe for making bread requires 3^2 cups of flour. If you want to make 2^2 times the amount of bread, how many cups of flour will you need?
(A) 3^2
(B) 3^3
(C) 3^4
(D) $2^2 \cdot 3^2$
(E) $2^3 \cdot 3^2$
- Q3.** A building is 2^3 meters tall. If a new building is 2^2 times taller, how tall is the new building?
(A) 2^3 meters
(B) 2^4 meters
(C) 2^5 meters
(D) 2^6 meters
(E) 2^7 meters
- Q4.** Simplify $(2^2)^3$
(A) 2^5
(B) 2^6
(C) 2^7
(D) 2^8
(E) 2^9
- Q5.** A chef needs to package 3^2 boxes of cookies. If each box can hold 2^3 cookies, how many cookies can be packaged in total?
(A) $2^3 \cdot 3^2$
(B) $2^2 \cdot 3^2$
(C) $2^4 \cdot 3^2$
(D) $2^5 \cdot 3^2$
(E) $2^6 \cdot 3^2$
- Q6.** Simplify $\frac{2^4}{2^2}$
(A) 2^2
(B) 2^3
(C) 2^4
(D) 2^5
(E) 2^6
- Q7.** A carpenter needs to make 2^3 sets of stairs. If each set requires 3^2 pieces of wood, how many pieces of wood will the carpenter need in total?
(A) $2^3 \cdot 3^2$
(B) $2^2 \cdot 3^2$
(C) $2^4 \cdot 3^2$
(D) $2^5 \cdot 3^2$
(E) $2^6 \cdot 3^2$
- Q8.** Simplify $(3^2)^2$
(A) 3^3
(B) 3^4
(C) 3^5
(D) 3^6
(E) 3^7

Open Ended Questions (FRQ)

- Q9.** A bakery is making a cake that requires 2^3 cups of sugar. If they want to make 2^2 times the amount of cake, and each cup of sugar costs 3^2 dollars, how much will the sugar cost in total? Show your work and simplify your answer.
- Q10.** A construction company is building a skyscraper that is 2^4 meters tall. If they want to add 2^3 more meters to the height, and each meter of the building requires 3^2 kilograms of steel, how many kilograms of steel will the company need in total? Show your work and simplify your answer.

Chef's Tips

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- Tip 1: Encourage students to use exponent rules to simplify expressions.
- Tip 2: Remind students to consider the context of the problem when applying exponent rules.
- Tip 3: Emphasize the importance of showing work and simplifying answers in free-response questions.